

Translated

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Prüfer code

In this article we will look at the so-called **Prüfer code** (or Prüfer sequence), which is a way of encoding a labeled tree into a sequence of numbers in a unique way.

With the help of the Prüfer code we will prove **Cayley's formula** (which specified the number of spanning trees in a complete graph). Also we show the solution to the problem of counting the number of ways of adding edges to a graph to make it connected.

Note, we will not consider trees consisting of a single vertex - this is a special case in which multiple statements clash.

Prüfer code

The Prüfer code is a way of encoding a labeled tree with n vertices using a sequence of $n - 2$ integers in the interval $[0; n-1]$. This encoding also acts as a **bijection** between all spanning trees of a complete graph and the numerical sequences.

Although using the Prüfer code for storing and operating on tree is impractical due the specification of the representation, the Prüfer codes are used frequently: mostly in solving combinatorial problems.

The inventor - Heinz Prüfer - proposed this code in 1918 as a proof for Cayley's formula.

Building the Prüfer code for a given tree

The Prüfer code is constructed as follows. We will repeat the following procedure $n - 2$ times: we select the leaf of the tree with the smallest number, remove it from the tree, and write down the number of the vertex that was connected to it. After $n - 2$ iterations there will only remain 2 vertices, and the algorithm ends.

Thus the Prüfer code for a given tree is a sequence of $n - 2$ numbers, where each number is the number of the connected vertex, i.e. this number is in the interval $[0, n-1]$.

The algorithm for computing the Prüfer code can be implemented easily with $O(n \log n)$ time complexity, simply by using a data structure to extract the minimum (for instance `set` or `priority_queue` in C++), which contains a list of all the current leaves.